



A Neuro-Symbolic Multi-Agent Framework for Pre-Release Audience Response Generation in Bangla Cinema

A Verifier-in-the-Loop Approach

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August 7, 2026

Background

01 Audience Commentary

Audience comments help explain how viewers interpret a film's characters, events, and narrative. However, authentic responses to a film become available only after audiences have watched it.

02 Pre-release Generation

Before release, LLMs can generate plausible, audience-like responses in Bangla from a film synopsis. This creates an opportunity to study pre-release response generation without claiming access to actual audience opinions.

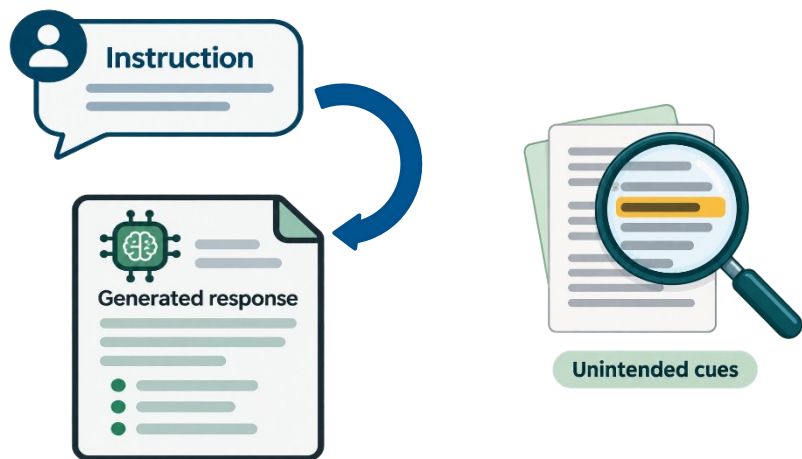
03 Generation and Control

Fluent generation alone does not ensure that a response consistently expresses a requested characteristic. It also does not establish that the generated text represents genuine audience reception, preference, or behavior.

04 Bangla Research Context

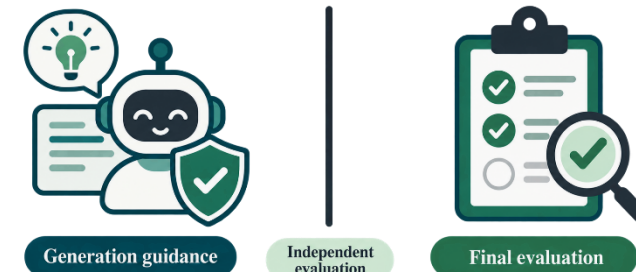
This study uses a 5,000 raw Bangla movie-review corpus to examine textual response patterns. Since the corpus contains no movie-title or audience-demographic information, it supports controlled response generation, but not film-specific audience modelling or audience-group prediction.

Motivation



2. Control May Depend on Unintended Cues

A generated response may satisfy a classifier because of superficial features, such as response length, rather than the intended response characteristics.



4. Evaluator Separation

Final outcomes require an evaluator outside the generation loop.

1. Prompting Does Not Guarantee Control

An LLM may be instructed to produce a general or plot-specific response, but the instruction alone cannot confirm that the requested distinction has been achieved.



3. Unreliable Self-correction

Revision may fail without external, task-specific feedback.

Research Need

To make Bangla cinema response generation controllable and auditable through external verification, bounded revision, and independent outcome evaluation.

Problem Statement

Central Research Problem

Prompting alone cannot ensure that a short Bangla cinema response follows the requested engagement-specificity level or that apparent improvement extends beyond the verifier-guided generation.

Connected Problem Dimensions



1. Construct Ambiguity

The target distinction may reflect sentiment, text length, or corpus artefacts rather than engagement specificity.



2. Control Uncertainty

Prompting alone cannot ensure that a response follows the requested engagement-specificity level.



3. Evaluation Circularity

The verifier guiding generation cannot independently certify the resulting improvement.

Research Aim and Objectives

Aim

To design and evaluate an auditable framework for controllable Bangla cinema response generation.

	Examine the Corpus	01		Validate the Construct	02		Isolate Evaluation	03		Build the Framework	04		Evaluate the Strategies	05
IDENTIFY THE CONSTRUCT			ESTABLISH HUMAN RECOGNIZABILITY			PREVENT EVALUATION CIRCULARITY			ENABLE CONTROLLED GENERATION			ASSESS PERFORMANCE AND RISK		
Identify a meaningful response distinction while testing potential confounds.			Determine whether native Bangla readers can recognize the distinction.			Separate generation guidance from final outcome assessment.			Implement a bounded and traceable neuro-symbolic multi-agent workflow.			Assess controllability, human recoverability, computational cost, and evaluator divergence.		

Research Questions

RQ1

Can a meaningful, stable, and human-recognizable response distinction be recovered from unlabeled Bangla reviews?

RQ2

To what extent do the registered generation strategies improve target-level controllability over zero-shot generation in Bangla?

RQ3

How does symbolic information function as acceptance gating versus diagnostic feedback in verifier-guided generation?

RQ4

Does iterative optimization against Verifier-A create measurable divergence from the independent Verifier-B?

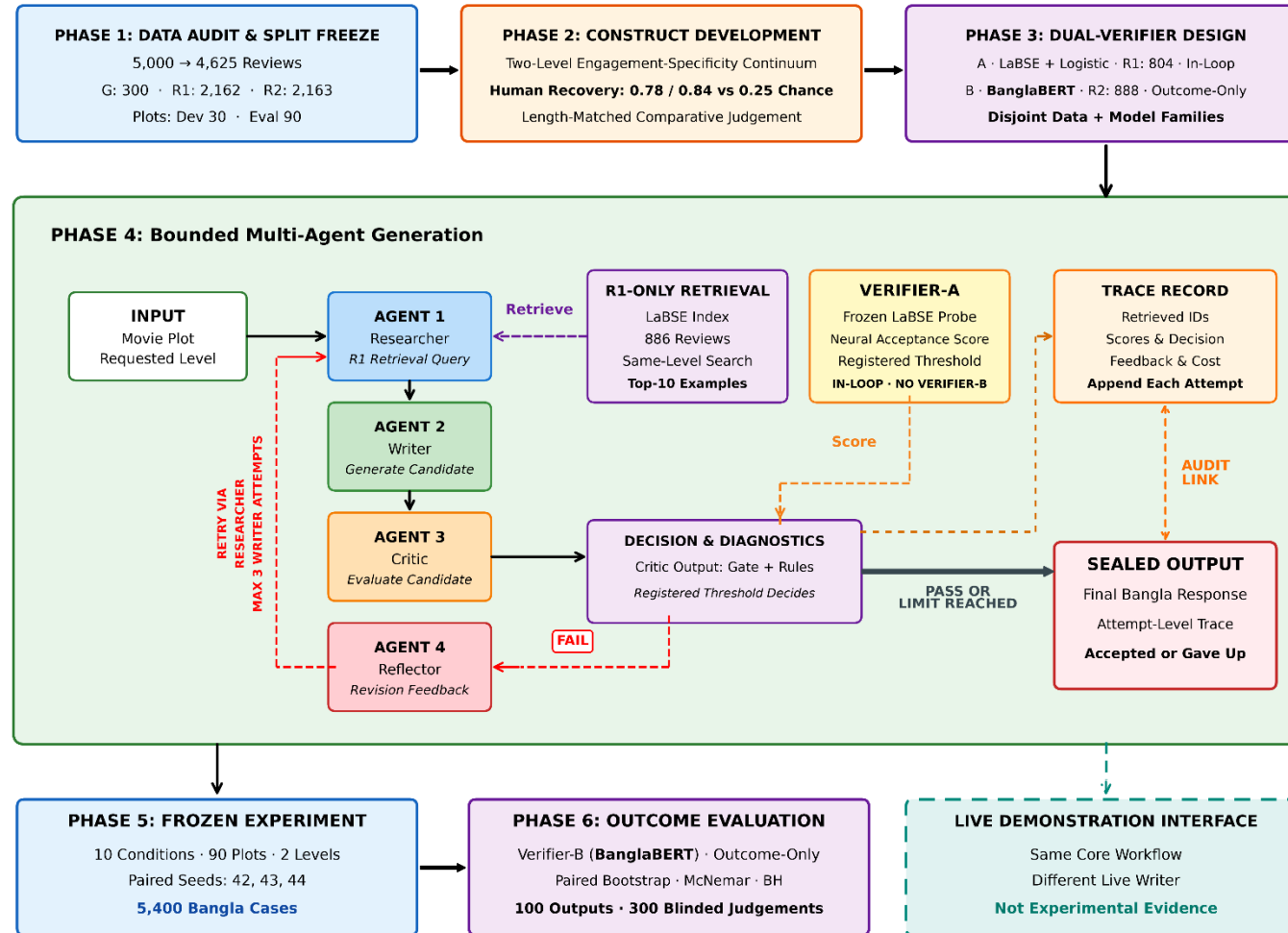
Related Work and Research Gap

Approach	Task / Language	Retrieval	Revision Signal	Symbolic Diagnosis	Outcome Scorer Isolated?	Human / Outcome Evaluation
MoP	Population-conditioned text generation	—	Persona conditioning	—	—	Study-specific evaluation
FUDGE	Controlled text generation	—	Learned discriminator during decoding	—	—	Task metrics
Self-Refine	Multi-task refinement	—	Same-model self-feedback	—	—	Task-dependent human and automatic evaluation
HybridRAG-BN	Bangla knowledge-base QA	BM25 + BGE-M3	Fine-tuned verifier and refiner	—	—	Competition token-F1
SymDiag	Multi-step reasoning	—	Symbolic diagnosis supports repair	Yes	No isolated A/B wall	Manually audited diagnosis
Present Study	Bangla cinema-response generation	R1 level-specific exemplars	Verifier-A + bounded feedback	Deterministic diagnostic rules	Verifier-B outside the loop	Construct study + blinded output study

Research Gap

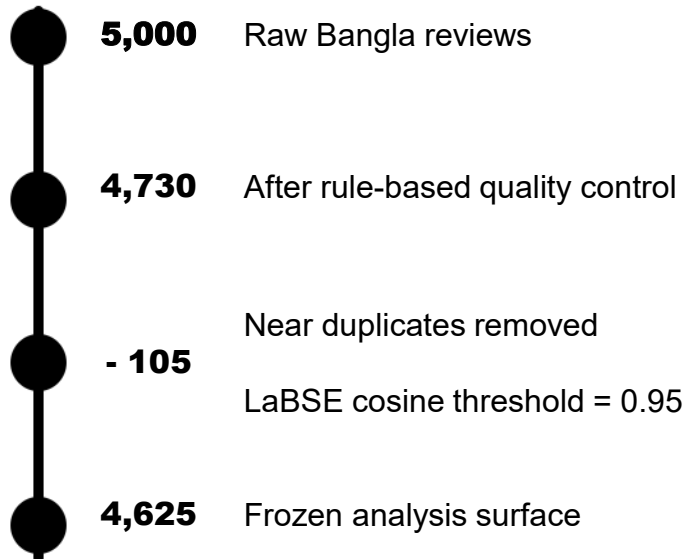
Among the studies reviewed, these components have not been jointly evaluated for Bangla cinema-response generation with verifier isolation and blinded human validation.

Research Methodology



Data Preparation and Axis Discovery

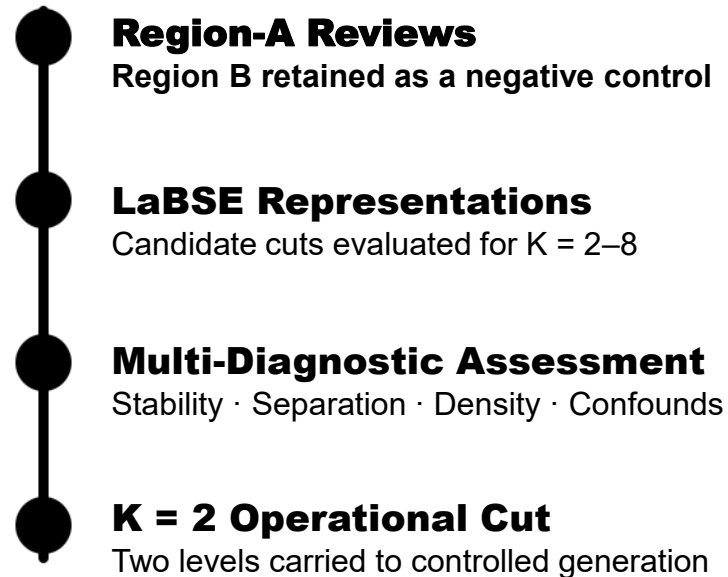
01 DATA PREPARATION



Frozen Splits

Gold-300: 300 · R1: 2,162 · R2: 2,163

02 AXIS DISCOVERY



Continuum cut—not natural groups

03 EVIDENCE & INTERPRETATION

Reproducible Under the Registered Rule

Prediction strength: 0.8605
Pre-specified cutoff: ≥ 0.80
Bootstrap ARI: 0.9399 ± 0.0290

Not Discrete

Silhouette: 0.0534
Gap statistic: No K selected
HDBSCAN: 100% noise

Human-Recognizable

Intrusion accuracy: 0.780 / 0.840
Direction judgment: 0.850
Length heuristic: 0.160

Dual-Verifier Design

IN-LOOP CONTROLLER

Verifier-A

Frozen LaBSE encoder + L2-regularised logistic head

- Trained on 804 labelled R1 reviews
- Scores drafts inside the generation loop
- Supports acceptance and bounded retry
- Selects candidates only where registered

PRIVILEGE
ISOLATION

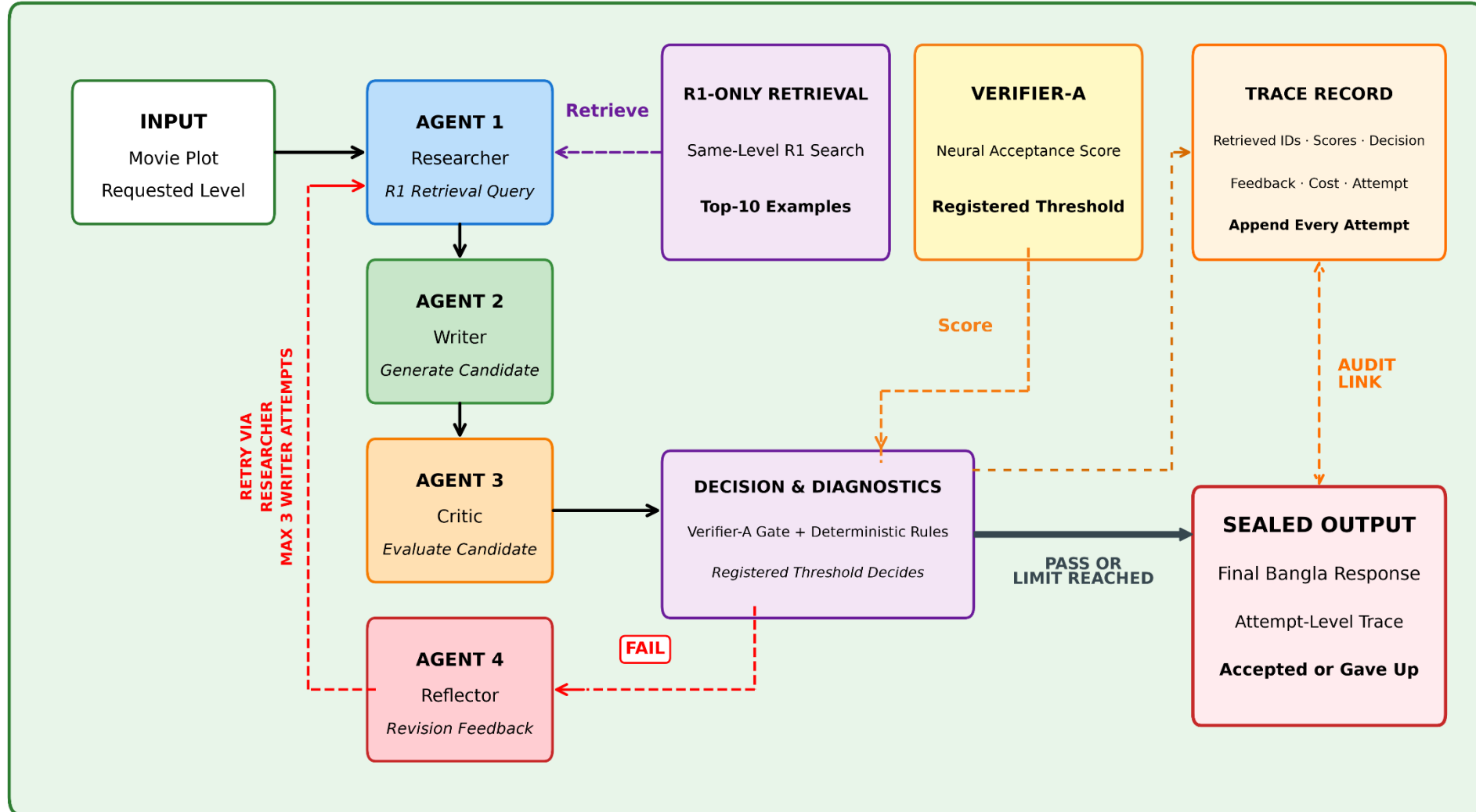
OUTCOME EVALUATOR

Verifier-B

BanglaBERT · fine-tuned from scratch

- Trained on 888 disjoint R2 reviews
- Scores sealed outputs after generation
- Has no access to retrieval or prompts
- Cannot provide feedback or trigger regeneration

Bounded Neuro-Symbolic Generation Loop

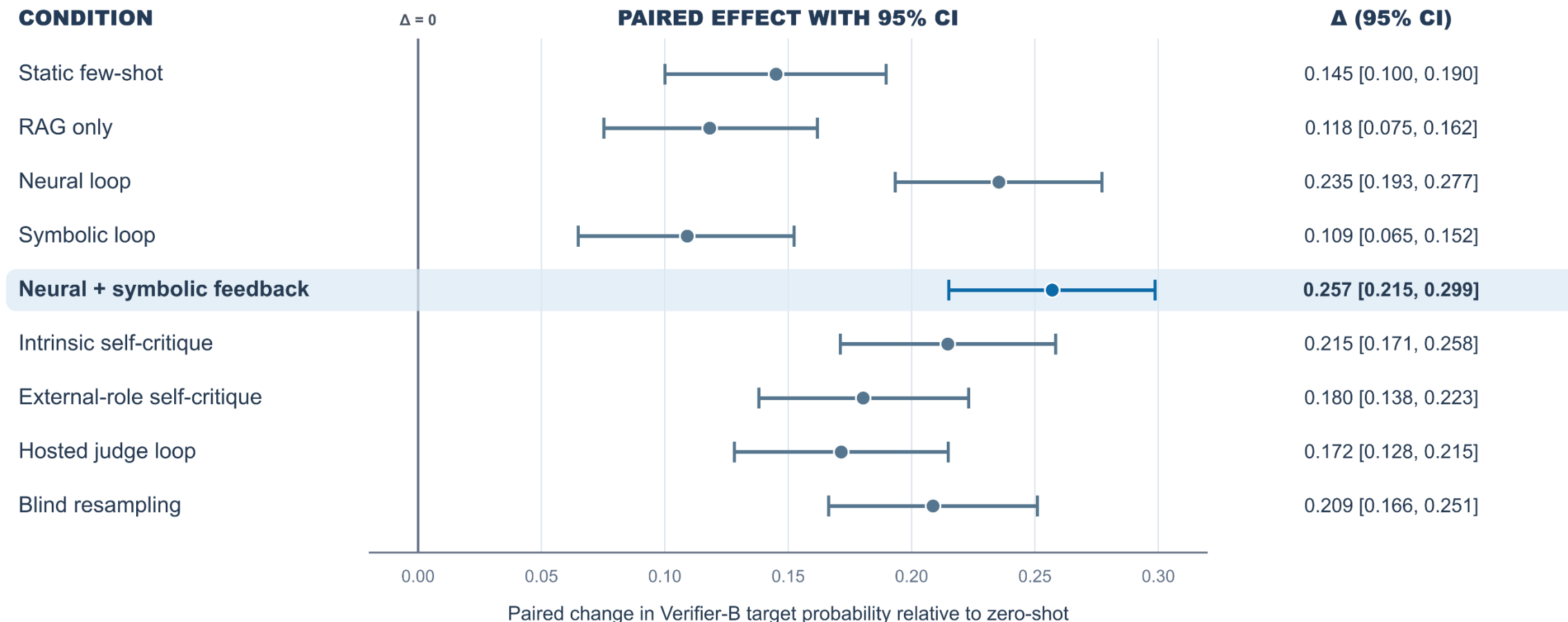


Experimental Design

Design Role	Conditions	What the Comparison Tests?	Evaluation Surface	
Reference	Zero-shot	Performance without examples or generation control	90	Held-out plots
Example Conditioning	- Static few-shot - RAG-only	Random examples versus level-specific retrieval	2	Requested axis levels
Gated Revision	- Neural gate - Symbolic gate - Neural gate + symbolic feedback	Acceptance control and the role of diagnostic feedback	10	Experimental conditions
Alternative Controls	- Intrinsic self-critique - External-role self-critique - Hosted-judge loop - Budget-matched blind resampling	Revision versus critique, stronger judging and additional sampling	3	Paired replicates
			5,400	Bangla cases

Outcome	Pairing	Confirmatory Family	Inference
Verifier-B target probability	Plot × level × replicate	9 active conditions vs zero-shot	Bootstrap · McNemar · BH

Experimental Results



All nine preregistered contrasts were positive, with 95% paired-bootstrap intervals excluding zero.

Human Evaluation

3 native Bangla-speaking annotators • 100 blinded outputs • 300 judgments

POOLED HUMAN JUDGMENTS

91.3%

Requested-level accuracy

95% CI [86.7%, 95.7%]

INTER-RATER AGREEMENT

Krippendorff's $\alpha = 0.841$

95% CI [0.747, 0.920]

Instrument	Level 0	Level 1	L1 – L0
Human majority	0.92	0.92	0.00
Verifier-A	0.70	0.84	+0.14
Verifier-B	0.60	0.90	+0.30

Diagnostic Findings

Symbolic Roles

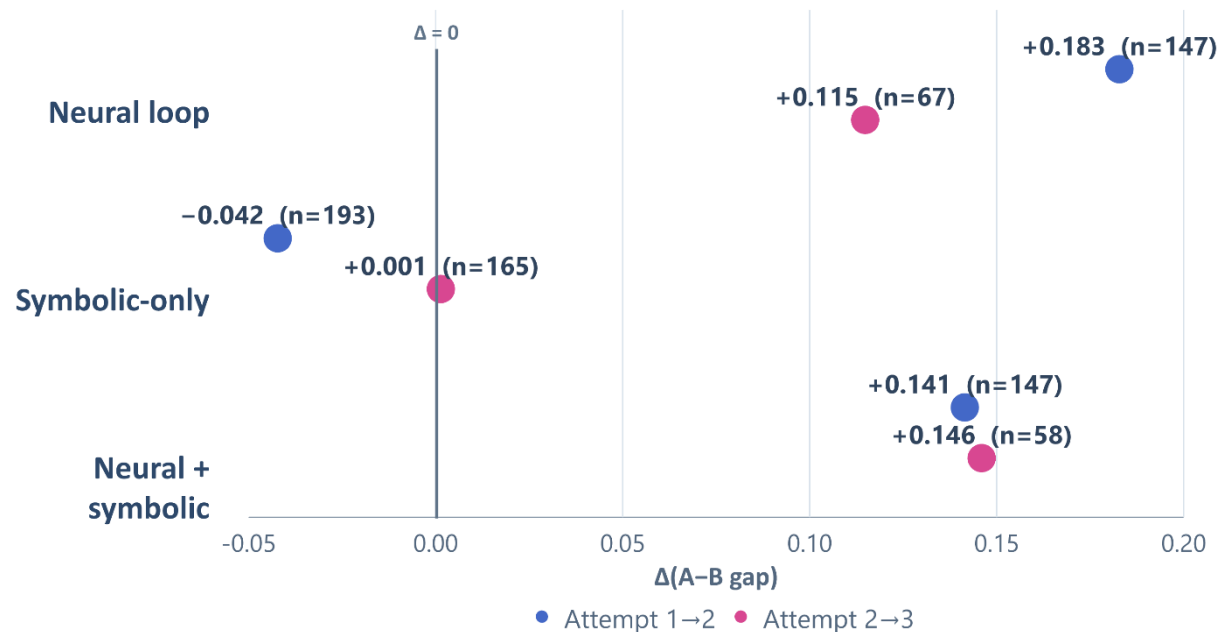
SYMBOLIC-ONLY ACCEPTANCE GATE

- Level-0 accuracy: **0.5185**
- Level-1 give-up rate: **53.33%**

HYBRID – NEURAL-ONLY

Δ target probability: +0.02159
 Δ binary accuracy: +0.02037
Exact McNemar: $p = 0.11728$
No statistically significant difference

Verifier Divergence



Research Contributions

1	Verifier Isolation	Separated the in-loop Verifier-A from the held-out Verifier-B, making proxy divergence observable.
2	Bounded and Auditable Generation	Developed a Researcher–Writer–Critic–Reflector workflow with bounded revision, recorded compute, and attempt-level traces.
3	Construct and Measurement Validation	Rejected the unsupported persona interpretation and retained a human-recognizable two-level operational distinction—not natural audience clusters.
4	Blinded Human Evidence	Demonstrated requested-level recoverability through 300 blinded judgments from native Bangla-speaking annotators.
5	Functional Role of Symbolic Information	Distinguished symbolic diagnosis from adjudication: symbolic-only gating was weak, while diagnostic feedback provided an interpretable correction mechanism.

Limitations and Future Work

LIMITATIONS

1. **Limited Human Sample:** Three native Bangla-speaking annotators assessed 100 outputs, producing 300 judgments.
2. **Length–Level Entanglement:** Generated response length remains associated with the requested engagement-specificity level.
3. **Proxy-Based Evaluation:** Verifier-B is an isolated outcome measure—not human ground truth.
4. **Unconfirmed Symbolic Advantage:** The hybrid loop was not shown to outperform the neural-only loop in a confirmatory test.
5. **No Film-Linked Ground Truth:** The corpus contains no film identifiers or real pre-release audience responses.

FUTURE WORK

1. Recruit a larger and more diverse human sample with prospective ethics documentation.
2. Develop length-neutral controls and test whether the requested distinction remains recoverable.
3. Stress-test both verifiers and validate their scores against human judgments.
4. Preregister a powered direct comparison between the neural-only and hybrid loops.
5. Collect film-linked audience data before making any audience-prediction claim.

Conclusion

- **Construct:** A reproducible Region-A continuum cut was human-recognizable under length-matched comparative judgment, but it was not a natural audience cluster and generated length remained entangled with the requested level.
- **Controlled generation:** All nine registered alternatives increased sealed Verifier-B target probability relative to zero-shot; on the balanced 100-item subset, readers achieved 91.3% target-level match.
- **Symbolic role:** Symbolic-only acceptance gating was weak. Symbolic rules were more defensible as diagnostic revision guidance, but hybrid superiority over the neural-only loop was not established.
- **Verifier risk:** In the two neural-gated loops, Verifier-A–Verifier-B gaps widened across revisions, consistent with proxy overoptimization—not evidence of declining human-perceived quality.

**THANK
YOU!**

